

The decreasing root projections in the CAR completion

MathlibAnnex.CStarAlgebra.CAR.rootFlag

def

The distinguished stage corners give a nested sequence of projections supporting the root state.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. Let ω be the product-vector state determined by $\omega(\iota_n(a)) = a_{00}$. Define $f_n = \iota_n(E_{00}) = e_{00}^{(n)}$. This sequence is the root flag.

Definition

At stage n , the matrix E_{00} has a single nonzero entry, equal to 1 at the distinguished diagonal position. Send it into A by ι_n . At the next stage, the new distinguished projection lies under the image of the preceding one, giving the decreasing relation. The finite matrix identity $E_{00}aE_{00} = a_{00}E_{00}$, together with the coordinate-preserving embeddings, explains the compression formula at later stages.

Assumptions

The zero coordinate is the same distinguished coordinate used by the standard stage inclusions. The index n ranges over all natural numbers.

Conclusion

Each f_n is a self-adjoint projection, $f_0 = 1$, $\omega(f_n) = 1$, and $f_{n+1} \leq f_n$. For an element from stage m , compression by f_n at any later stage $n \geq m$ is exactly scalar: $f_n \iota_m(a) f_n = a_{00} f_n$. The cited results establish these properties of the defined sequence.

The term rank one refers to E_{00} inside the finite matrix algebra. It does not assert that the image of f_n has one-dimensional range in a representation of the completed algebra. No norm or strong limit of the flag is part of this definition.

Main citations

- Definition and its exact construction — MathlibAnnex.CStarAlgebra.CAR.rootFlag
- Finite root projection — MathlibAnnex.CStarAlgebra.CAR.rootProjection
- Finite root compression — MathlibAnnex.CStarAlgebra.CAR.rootProjection_mul_mul
- Initial root projection — MathlibAnnex.CStarAlgebra.CAR.rootFlag_zero
- Projection property — MathlibAnnex.CStarAlgebra.CAR.isStarProjection_rootFlag

- State support — `MathlibAnnex.CStarAlgebra.CAR.rootState_rootFlag`
- Decreasing root flag — `MathlibAnnex.CStarAlgebra.CAR.antitone_rootFlag`
- Compression at later stages — `MathlibAnnex.CStarAlgebra.CAR.rootFlag_mul_ofStage_mul`

Lean source signature (exact)

```
noncomputable def rootFlag (n : ℕ) : Limit :=
  ofStage n (rootProjection n)
```

In the source	Mathematical meaning
<code>n : ℕ; Limit</code>	Every stage index $n \geq 0$ and the completed CAR algebra A .
<code>rootProjection n</code>	The distinguished diagonal matrix unit $E_{00} \in A_n = M_{2^n}(\mathbb{C})$.
<code>ofStage n (rootProjection n)</code>	Apply the canonical stage inclusion to that projection: $f_n = \iota_n(E_{00}) \in A$. This is the entire defining RHS.
<code>rootFlag n</code>	The projection f_n itself, not the difference shell $f_n - f_{n+1}$. Its decreasing order and compression properties are separately proved facts.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.rootFlag`
 Accepted content SHA-256: `ee5411e794a6cfd2e57ba8e51113f8d232a4d683926984d71c4bc9d54c445ab`
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 Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`